

DOM = HTML ka JavaScript ke andar bana hua object/tree representation.

JavaScript directly `.html` file ko manipulate nahi kar rahi. Browser HTML ko parse karke **DOM tree** banata hai, aur JavaScript us DOM tree ko manipulate karti hai.

1. Sabse pehle HTML → DOM kaise banta hai?

Tumhara HTML:

```
HTML

<body>

  <h1 id="h1">h1 heading</h1>

  <h2 id="h2">h2 heading</h2>

  <h3 class="h3">h3 heading</h3>

  <h4 class="h4">h4 heading</h4>
```

Js dom manipulation

HTML

```
<h4 class="h4">h4 heading</h4>
```

```
<h5 class="h5">h5 heading</h5>
```

```
<h6>h6 heading</h6>
```

```
<p>paragraph hai</p>
```

```
<div class="box1">
```

```
  box1
```

```
  <ul>
```

```
    <li>mango</li>
```

```
    <li>orange</li>
```

```
    <li>banana</li>
```

```
  </ul>
```

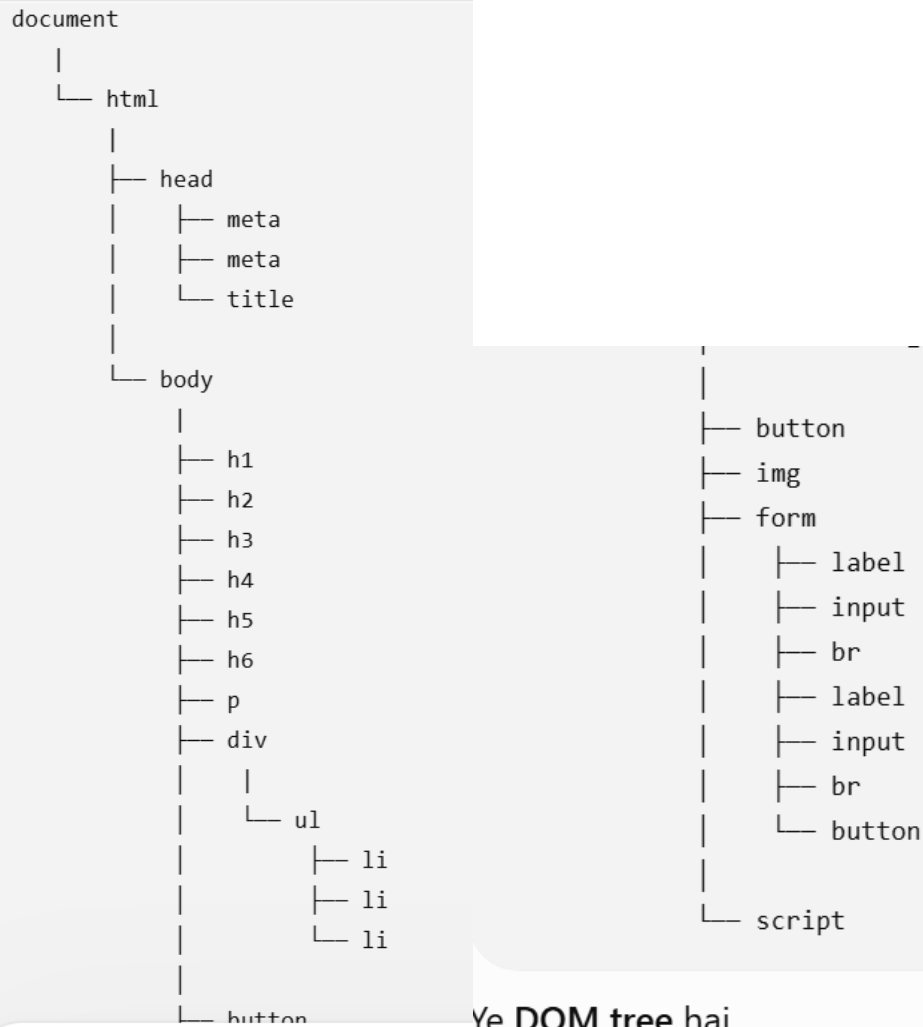
```
</div>
```

```
<button>click me</button>
```



Browser isko roughly tree me convert karta hai:

Js dom manipulation



Ye DOM tree hai.

`window` browser ki **top-level object** hai.

Iske andar browser environment ki bahut saari cheezein hoti hain:

</> JavaScript

```

window.document
window.location
window.history
window.localStorage
window.setTimeout
window.alert

```

Tum:

```
</> JavaScript
```

```
console.log(location.href);
```

karoge to complete URL milega.

Aur:

```
</> JavaScript
```

```
console.log(location.hostname);
```

hostname.

```
</> JavaScript
```

```
console.log(location.pathname);
```

pathname.

6. window.localStorage

```
</> JavaScript  
console.log(window.localStorage);
```

Browser ke local storage ko access karta hai.

Example:

```
</> JavaScript  
localStorage.setItem("name", "Tarun");
```

Read:

```
</> JavaScript  
console.log(localStorage.getItem("name"));
```

24. Selection methods ek baar clearly

Method	Selector	Result
<code>getElementById()</code>	<code>"h1"</code>	one element
<code>querySelector()</code>	<code>"#h1"</code>	first matching
<code>querySelectorAll()</code>	<code>".h3"</code>	all matching
<code>getElementsByClassName()</code>	<code>"h3"</code>	collection
<code>getElementsByTagName()</code>	<code>"p"</code>	collection

Js dom manipulation

```
// input select
let input = document.querySelector("#username");

// input ka placeholder
input.placeholder = "Enter fruit";

// button select
let btn = document.getElementById("btn");

// button ka text
btn.innerText = "Change Text";

// add button select
let addBtn = document.getElementById("add");

// add button ka text
addBtn.innerText = "Add Fruit";

// remove button select
let removeBtn = document.querySelector("#remove");
```



Js dom manipulation

```
let removeBtn = document.querySelector("#remove");

// remove button ka text
removeBtn.innerText = "Remove Fruit";

// =====
// STYLE
// =====

// h1 select
let h1Style = document.querySelector("#h1");

// h1 color
h1Style.style.color = "red";

// h2 select
let h2Style = document.getElementById("h2");
```



Js dom manipulation

```
let h2Style = document.getElementById("h2");

// h2 background
h2Style.style.backgroundColor = "yellow";

// h3 select
let h3Style = document.querySelector(".h3");

// h3 font size
h3Style.style.fontSize = "30px";

// box select
let boxStyle = document.querySelector(".box1");

// box background
boxStyle.style.backgroundColor = "lightblue";
```

Js dom manipulation

```
// -----  
// TRAVERSING  
// =====  
  
// ul select  
let ulParent = document.getElementById("fruitList");  
  
// parent element  
console.log(ulParent.parentElement);  
  
// ul select  
let ulChildren = document.querySelector("#fruitList");  
  
// children  
console.log(ulChildren.children);  
  
// ul select  
let ulFirst = document.getElementById("fruitList");  
  
// first child
```



Js dom manipulation

```
// ul select
let ulFirst = document.getElementById("fruitList");

// first child
console.log(ulFirst.firstChild);

// ul select
let ulLast = document.querySelector("#fruitList");

// last child
console.log(ulLast.lastElementChild);

// =====
// NEXT / PREVIOUS
// =====

// ul select
let firstLi = document.querySelector("#fruitList");
```



```
let firstLi = document.querySelector("#fruitList");

// first li
firstLi = firstLi.firstChild;

// next li
console.log(firstLi.nextElementSibling);

// ul select
let lastLi = document.getElementById("fruitList");

// last li
lastLi = lastLi.lastElementChild;

// previous li
console.log(lastLi.previousElementSibling);
```

Js dom manipulation

```
// saare li select
let fruits = document.querySelectorAll("#fruitList li");

// NodeList
console.log(fruits);

// har fruit
fruits.forEach(function(fruit) {

    console.log(fruit.innerText);

});
```

```
// class ke elements
let headingClass =
    document.getElementsByClassName("h3");

// HTMLCollection
console.log(headingClass);

// first element
console.log(headingClass[0]);

// =====
// GET ELEMENTS BY TAG NAME
// =====

// saare li
let allLi = document.getElementsByTagName("li");

// HTMLCollection
console.log(allLi);
```



Js dom manipulation

```
// first li
console.log(allLi[0]);

// second li
console.log(allLi[1]);

// third li
console.log(allLi[2]);

// =====
// CLICK EVENT
// =====

// button select
let clickBtn = document.getElementById("btn");
```

Js dom manipulation

```
// click event
clickBtn.addEventListener("click", function() {

    // paragraph select
    let para = document.getElementById("para");

    // paragraph change
    para.innerText = "Button click hua";

});

// =====
// INPUT EVENT
// =====

// input select
let userInput = document.querySelector("#username");
```

```
// input event
userInput.addEventListener("input", function() {

    // input ki value
    console.log(userInput.value);

});

// =====
// CREATE ELEMENT
// =====

// add button select
let addButton = document.getElementById("add");

// click event
addButton.addEventListener("click", function() {
```

</> JavaScript

```
// click event
addButton.addEventListener("click", function() {

    // input select
    let inputBox = document.querySelector("#username");

    // input value
    let value = inputBox.value;

    // empty hai
    if (value === "") {
        return;
    }

    // new li create
    let li = document.createElement("li");

    // li ka text
    li.innerText = value;
```



Js dom manipulation

```
let h2Style = document.getElementById("h2");

// h2 background
h2Style.style.backgroundColor = "yellow";

// h3 select
let h3Style = document.querySelector(".h3");

// h3 font size
h3Style.style.fontSize = "30px";

// box select
let boxStyle = document.querySelector(".box1");

// box background
boxStyle.style.backgroundColor = "lightblue";
```

Js dom manipulation

```
// -----  
// TRAVERSING  
// =====  
  
// ul select  
let ulParent = document.getElementById("fruitList");  
  
// parent element  
console.log(ulParent.parentElement);  
  
// ul select  
let ulChildren = document.querySelector("#fruitList");  
  
// children  
console.log(ulChildren.children);  
  
// ul select  
let ulFirst = document.getElementById("fruitList");  
  
// first child
```



Js dom manipulation

```
// ul select
let ulFirst = document.getElementById("fruitList");

// first child
console.log(ulFirst.firstChild);

// ul select
let ulLast = document.querySelector("#fruitList");

// last child
console.log(ulLast.lastElementChild);

// =====
// NEXT / PREVIOUS
// =====

// ul select
let firstLi = document.querySelector("#fruitList");
```



Js dom manipulation

```
let firstLi = document.querySelector("#fruitList");

// first li
firstLi = firstLi.firstChild;

// next li
console.log(firstLi.nextElementSibling);

// ul select
let lastLi = document.getElementById("fruitList");

// last li
lastLi = lastLi.lastElementChild;

// previous li
console.log(lastLi.previousElementSibling);
```

```
// saare li select
let fruits = document.querySelectorAll("#fruitList li");

// NodeList
console.log(fruits);

// har fruit
fruits.forEach(function(fruit) {

    console.log(fruit.innerText);

});
```

Js dom manipulation

```
// class ke elements
let headingClass =
    document.getElementsByClassName("h3");

// HTMLCollection
console.log(headingClass);

// first element
console.log(headingClass[0]);

// =====
// GET ELEMENTS BY TAG NAME
// =====

// saare li
let allLi = document.getElementsByTagName("li");

// HTMLCollection
console.log(allLi);
```



```
// first li
console.log(allLi[0]);

// second li
console.log(allLi[1]);

// third li
console.log(allLi[2]);

// =====
// CLICK EVENT
// =====

// button select
let clickBtn = document.getElementById("btn");
```

Js dom manipulation

```
// click event
clickBtn.addEventListener("click", function() {

    // paragraph select
    let para = document.getElementById("para");

    // paragraph change
    para.innerText = "Button click hua";

});

// =====
// INPUT EVENT
// =====

// input select
let userInput = document.querySelector("#username");
```

Js dom manipulation

```
// input event
userInput.addEventListener("input", function() {

    // input ki value
    console.log(userInput.value);

});

// =====
// CREATE ELEMENT
// =====

// add button select
let addButton = document.getElementById("add");

// click event
addButton.addEventListener("click", function() {
```

</> JavaScript

```
// click event
addButton.addEventListener("click", function() {

    // input select
    let inputBox = document.querySelector("#username");

    // input value
    let value = inputBox.value;

    // empty hai
    if (value === "") {
        return;
    }

    // new li create
    let li = document.createElement("li");

    // li ka text
    li.innerText = value;
```



Js dom manipulation

```
let fruitList = document.getElementById("fruitList");

// li ko ul me add
fruitList.appendChild(li);

// input clear
inputBox.value = "";

});

// =====
// REMOVE ELEMENT
// =====

// remove button select
let removeButton =
  document.querySelector("#remove");
```



Js dom manipulation

```
// click event
removeButton.addEventListener("click", function() {

    // ul select
    let fruitList = document.getElementById("fruitList");

    // first li
    let firstFruit = fruitList.firstChild;

    // agar element hai
    if (firstFruit) {

        // remove
        firstFruit.remove();

    }

}):
```

Js dom manipulation

```
DOM
|
├─ SELECT
|   ├── querySelector()
|   ├── querySelectorAll()
|   ├── getElementById()
|   ├── getElementsByClassName()
|   └─ getElementsByTagName()
|
├─ INFORMATION
|   ├── tagName
|   ├── nodeName
|   ├── nodeType
|   ├── textContent
|   ├── innerText
|   ├── innerHTML
|   ├── outerHTML
|   └─ outerText
|
├─ ATTRIBUTE
|   ├── getAttribute()
|   ├── setAttribute()
|   ├── removeAttribute()
|   ├── hasAttribute()
|   └─ getAttributeNames()
```

```
|  
|— CLASS  
|   |— classList.add()  
|   |— classList.remove()  
|   |— classList.toggle()  
|   └─ classList.contains()  
|  
|— TRAVERSING  
|   |— parentElement  
|   |— children  
|   |— childNodes  
|   |— firstElementChild  
|   |— lastElementChild  
|   |— nextElementsSibling  
|   └─ previousElementsSibling  
|  
|— CREATE  
|   |— createElement()  
|   |— append()  
|   |— appendChild()  
|   |— prepend()  
|   |— before()  
|   |— after()  
|   |— insertAdjacentElement()  
|   └─ cloneNode()
```

```
.
├── DELETE
│   ├── remove()
│   └── removeChild()
└── EVENTS
    ├── onclick
    ├── addEventListener()
    ├── click
    ├── dblclick
    ├── mouseover
    ├── mouseenter
    ├── mouseleave
    ├── keydown
    ├── keyup
    ├── input
    ├── focus
    ├── blur
    └── contextmenu
```

```
<body>

  <h1 id="h1">H1 Heading</h1>

  <h2 id="h2">H2 Heading</h2>

  <h3 class="h3">H3 Heading</h3>

  <h4 class="h4">H4 Heading</h4>

  <p id="para">Old paragraph</p>

  
```

```
<div class="box1">

  <h3>Fruit List</h3>

  <ul id="fruitList">

    <li>Mango</li>
    <li>Orange</li>
    <li>Banana</li>

  </ul>

</div>

<input
  id="username"
  type="text"
  placeholder="Enter your name"
```

```
>  
  
<br><br>  
  
<button id="btn">Click Me</button>  
  
<button id="addBtn">Add Fruit</button>  
  
<button id="removeBtn">Remove Fruit</button>  
  
<button id="classBtn">Class Test</button>  
  
<script src="my.js"></script>  
  
</body>  
  
</html>
```

```
let h1 = document.querySelector("#h1");

console.log(h1);

console.log(h1.outerHTML);

console.log(h1.outerText);

console.log(h1.tagName);

console.log(h1.textContent);

console.log(h1.innerText);

console.log(h1.innerHTML);

h1.innerText = "DOM Master";

h1.style.color = "red";

console.log(h1.id);
```

```
console.log(h2.innerText);

console.log(h2.innerHTML);

h2.textContent = "JavaScript DOM";

h2.setAttribute("id", "mynewid");

console.log(h2.getAttribute("id"));

console.log(h2.hasAttribute("id"));

console.log(h2.getAttributeNames());
```

```
// =====
// H3
// =====
```

```
let h3 = document.querySelector(".h3");

console.log(h3);
```

```
console.log(p.textContent);

p.innerText = "Hi I am Tarun";

p.style.fontSize = "20px";


// =====
// IMAGE
// =====

let img = document.querySelector("#myImg");

console.log(img);

console.log(img.outerHTML);

console.log(img.tagName);

console.log(img.getAttribute("src"));
```

```
console.log(img.getAttribute("src"));

console.log(img.getAttribute("alt"));

img.setAttribute(
  "alt",
  "New Image"
);

console.log(img.getAttribute("alt"));

img.src = "https://via.placeholder.com/200";

img.classList.add("red");

console.log(
  img.classList.contains("red")
);

img.classList.toggle("red");

img.classList.remove("red");
```



```
let box1 = document.querySelector(".box1");

console.log(box1);

console.log(box1.outerHTML);

console.log(box1.tagName);


// parent
console.log(box1.parentElement);

console.log(box1.parentNode);


// children
console.log(box1.children);

console.log(box1.childElementCount);
```

```
// childNodes  
console.log(box1.childNodes);  
  
console.log(box1.childNodes.length);  
  
// first child  
console.log(box1.firstChild);  
  
// last child  
console.log(box1.lastElementChild);  
  
// style  
box1.style.padding = "20px";  
  
box1.style.border = "2px solid black";
```

```
let ul = document.getElementById("fruitList");

console.log(ul);

console.log(ul.outerHTML);

console.log(ul.tagName);

// parent
console.log(ul.parentElement);

// children
console.log(ul.children);

// first
console.log(ul.firstElementChild);

// last
console.log(ul.lastElementChild);
```



```
// JavaScript
// first li
let firstLi = ul.firstElementChild;

console.log(firstLi);

console.log(firstLi.innerText);

console.log(firstLi.nextElementSibling);

// second li
let secondLi =
    firstLi.nextElementSibling;

console.log(secondLi);

console.log(secondLi.innerText);

console.log(
    secondLi.previousElementSibling
);
```



```
// third li
let thirdLi =
    ul.lastElementChild;

console.log(thirdLi);

console.log(thirdLi.innerText);


// =====
// QUERY SELECTOR ALL
// =====

let fruits =
    document.querySelectorAll("#fruitList li");

console.log(fruits);

console.log(fruits.length);
```

```
fruits.forEach(function(fruit) {  
  
    console.log(fruit.innerText);  
  
});  
  
  
// =====  
// GET ELEMENTS BY CLASS NAME  
// =====  
  
let classElements =  
    document.getElementsByClassName("h3");  
  
console.log(classElements);  
  
console.log(classElements.length);  
  
console.log(classElements[0]);
```



```
let allLi =  
    document.getElementsByTagName("li");  
  
console.log(allLi);  
  
console.log(allLi.length);  
  
console.log(allLi[0]);  
  
console.log(allLi[1]);  
  
console.log(allLi[2]);
```

`</>` JavaScript

```
let input =  
    document.querySelector("#username");  
  
console.log(input);  
  
console.log(input.outerHTML);  
  
console.log(input.tagName);  
  
console.log(input.value);  
  
console.log(input.placeholder);  
  
input.value = "Tarun";  
  
input.placeholder = "Enter fruit";
```

```
... javascript

let btn =
    document.getElementById("btn");

console.log(btn);

console.log(btn.outerHTML);

console.log(btn.innerText);

console.log(btn.tagName);


// =====
// CREATE ELEMENT
// =====

let btn1 =
    document.createElement("button");

console.log(btn1);

console.log(btn1.tagName);
```

```
btn1.innerText = "New Button";

btn1.id = "newButton";

document.body.appendChild(btn1);
```

```
// =====
// APPEND
// =====
```

```
let p1 =
    document.createElement("p");

console.log(p1);

p1.innerText =
    "Append se add hua";

box1.append(p1);
```

JavaScript

```
let p2 =  
    document.createElement("p");
```

```
console.log(p2);
```

```
p2.innerText =  
    "Prepend se add hua";
```

```
box1.prepend(p2);
```

```
// =====  
// BEFORE  
// =====
```

```
let beforeH =  
    document.createElement("h4");
```

```
console.log(beforeH);
```

```
beforeH.innerText =  
    "Before Heading";
```

```
let afterH =
    document.createElement("h4");

console.log(afterH);

afterH.innerText =
    "After Heading";

box1.after(afterH);


// =====
// INSERT ADJACENT ELEMENT
// =====

let adjacentBtn =
    document.createElement("button");

console.log(adjacentBtn);

adjacentBtn.innerText =
    "Adjacent Button";
```

```
p.insertAdjacentElement(  
    "beforebegin",  
    adjacentBtn  
);
```

```
// =====  
// INSERT ADJACENT HTML  
// =====
```

```
p.insertAdjacentHTML(  
    "afterend",  
    "<b>Extra HTML</b>"  
);
```

```
p.insertAdjacentText(  
    "beforebegin",  
    "Before paragraph "  
);  
  
// =====  
// CLASSLIST  
// =====  
  
console.log(h1.classList);  
  
h1.classList.add("green");  
  
console.log(  
    h1.classList.contains("green")  
);  
  
h1.classList.toggle("active");  
  
h1.classList.remove("green");
```

```
btn.addEventListener(  
  "click",  
  function(event) {  
  
    console.log(  
      "Button clicked"  
    );  
  
    console.log(event);  
  
    console.log(event.target);  
  
    console.log(event.currentTarget);  
  
    console.log(event.type);  
  
  }  
);
```

```
btn.onclick = function() {  
  
    console.log(  
        "onclick executed"  
    );  
  
};
```

```
// =====  
// MOUSEOVER  
// =====
```

```
btn.addEventListener(  
    "mouseover",  
    function(event) {  
  
        console.log(  
            "Mouse over"  
        );  
  
    });
```

```
btn.addEventListener(  
  "mouseover",  
  function(event) {  
  
    console.log(  
      "Mouse over"  
    );  
  
    console.log(  
      event.target  
    );  
  
    console.log(  
      event.clientX  
    );  
  
    console.log(  
      event.clientY  
    );  
  
    console.log(  
      event.type  
    );  
  }  
);
```

```
</> JavaScript
```

```
btn.addEventListener(  
  "mouseenter",  
  function() {  
  
    console.log(  
      "Mouse entered"  
    );  
  
  }  
);
```

```
// =====  
// MOUSELEAVE  
// =====
```

```
btn.addEventListener(  
  "mouseleave",  
  function() {  
  
    console.log(  
      "Mouse left"    );  
  
  }  
);
```

```
btn.addEventListener(  
  "mouseout",  
  function() {  
  
    console.log(  
      "Mouse out"  
    );  
  
  }  
);  
  
// =====  
// KEYDOWN  
// =====  
  
input.addEventListener(  
  "keydown",  
  function(event) {  
  
    console.log(  
      "Key pressed"    );  
  
  }  
);
```

```
    "keydown",  
    function(event) {  
  
        console.log(  
            "Key pressed"  
        );  
  
        console.log(event);  
  
        console.log(  
            event.key  
        );  
  
        console.log(  
            event.code  
        );  
  
    }  
);
```

Js dom manipulation

```
// =====  
  
// KEYUP  
  
// =====  
  
input.addEventListener(  
    "keyup",  
    function(event) {  
  
        console.log(  
            "Key released"  
        );  
  
    }  
);
```

</> JavaScript

```
// =====  
  
// INPUT EVENT  
  
// =====  
  
input.addEventListener(  
    "input",  
    function(event) {  
  
        console.log(  
            input.value  
        );  
  
    }  
);
```

```
// =====  
// FOCUS  
// =====  
  
input.addEventListener(  
  "focus",  
  function() {  
  
    console.log(  
      "Input focused"  
    );  
  
  }  
);
```

JavaScript

```
//  
// BLUR  
// =====  
  
input.addEventListener(  
  "blur",  
  function() {  
  
    console.log(  
      "Input blurred"  
    );  
  
  }  
);
```

Js dom manipulation

```
let addBtn =  
  document.querySelector("#addBtn");  
  
console.log(addBtn);  
  
addBtn.addEventListener(  
  "click",  
  function() {  
  
    let value =  
      input.value;  
  
    if (value === "") {  
  
      return;  
  
    }  
  }  
);
```

```
    let li =  
      document.createElement("li");  
  
    li.innerText = value;  
  
    ul.appendChild(li);  
  
    input.value = "";  
  }  
);
```

```
let removeBtn =  
    document.querySelector("#removeBtn");  
  
console.log(removeBtn);  
  
removeBtn.addEventListener(  
    "click",  
    function() {  
  
        let firstFruit =  
            ul.firstChild;  
  
        if (firstFruit) {  
  
            firstFruit.remove();  
  
        }  
  
    }  
);
```



```
// CLASS BUTTON
// =====

let classBtn =
  document.querySelector("#classBtn");

console.log(classBtn);

classBtn.addEventListener(
  "click",
  function() {

    box1.classList.toggle(
      "active"
    );

    console.log(
      box1.classList.contains(
        "active"
      )
    )
  }
);
```



Js dom manipulation

```
// -----  
// CLONE NODE  
// =====  
  
let h1Clone =  
    h1.cloneNode(true);  
  
console.log(h1Clone);  
  
document.body.appendChild(  
    h1Clone  
);
```

```
// -----  
// REMOVE CHILD  
// =====  
  
let tempLi =  
    document.createElement("li");  
  
tempLi.innerText =  
    "Temporary Fruit";  
  
ul.appendChild(tempLi);  
  
ul.removeChild(tempLi);
```

```
// =====  
// REPLACE CHILD  
// =====  
  
let newLi =  
    document.createElement("li");  
  
newLi.innerText =  
    "New Fruit";  
  
let oldFruit =  
    ul.firstElementChild;  
  
ul.replaceChild(  
    newLi,  
    oldFruit  
);
```

```
... javascript

// =====
// DATASET
// =====

h1.dataset.user = "Tarun";

console.log(
  h1.dataset.user
);

// =====
// NODE INFORMATION
// =====

console.log(h1.nodeName);

console.log(h1.nodeType);

console.log(h1.nodeValue);
```

```
// DOCUMENT INFORMATION
// =====

console.log(document.title);

console.log(document.URL);

console.log(document.body);

console.log(document.head);


// =====
// WINDOW
// =====

console.log(window);

console.log(window.document);

console.log(window.location);
```